DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION
Year 6

HIGHER 2 GEOGRAPHY **9173/02**

INSERT

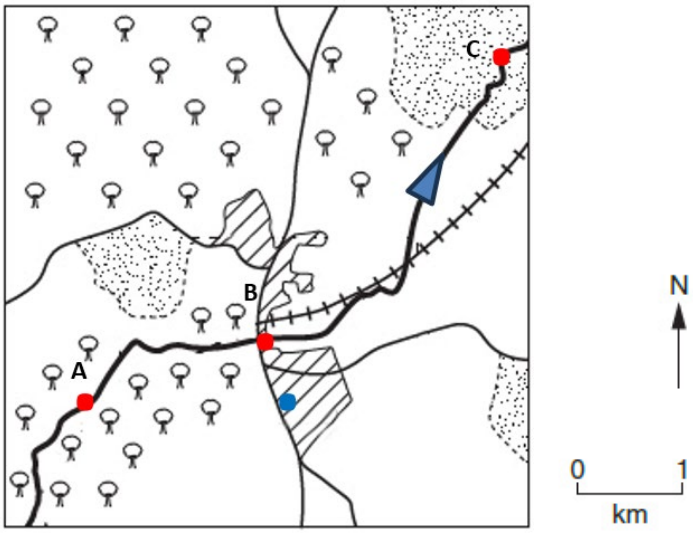
READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

BLANK PAGE

Resource 1A for Question 1

Map showing the fieldwork sites



Key

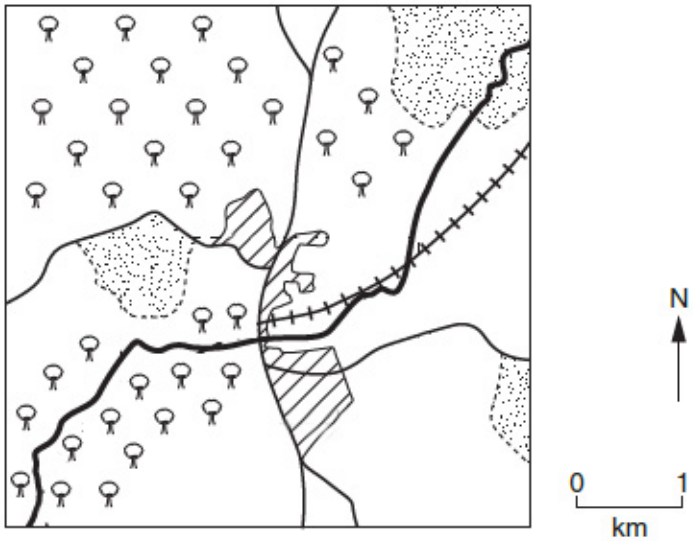
Fieldwork site

Direction of river flow

School

Resource 1B for Question 1

Land use near the river



Key

river

road

railway

woodland

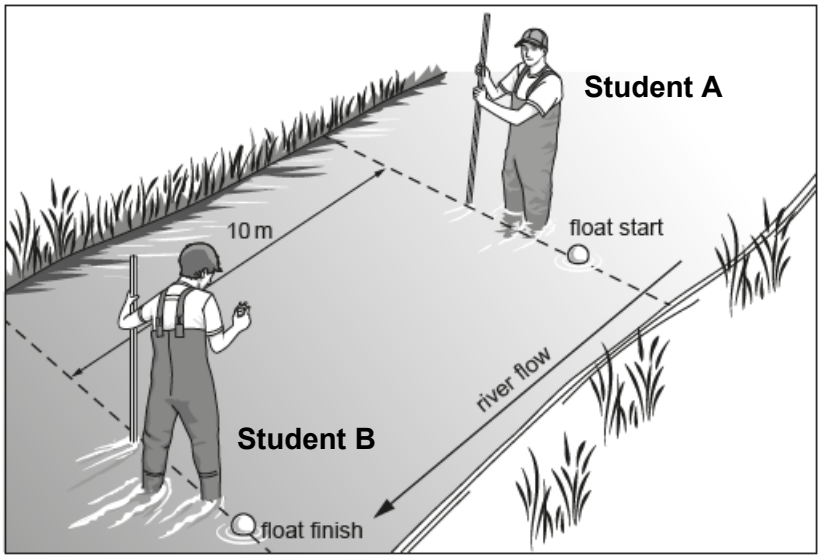
parkland

undeveloped land

urban area

Resource 2A for Question 1

Students using the float method to measure river velocity at each site



The float method measures the speed of a float that travels downstream over 10 metres.
Student B started the timer when Student A released the float. When it reached Student B, he stopped the timer.

Resource 2B for Question 1

Student using the flowmeter to measure river velocity at each site



Flowmeter measures river velocity by measuring the rotations of propeller and calculating the speed of the water.
The student took the reading by resting the flowmeter on the channel bed.

Resource 3A for Question 1

Results of velocity measurement at the three sites

Site	Method of measurement	Time of measurement	Velocity (m/sec)			
			Right side of channel	Centre of channel	Left side of channel	Average
A	Flowmeter	1.30pm	0.65	0.73	0.31	0.56
	Float method using apple	1.00pm	0.66	0.74	0.35	0.58
B	Flowmeter	2.30pm	0.54	0.37	0.48	0.46
	Float method using empty water bottle	1.00pm	0.61	0.82	0.52	0.65
C	Flowmeter	3.30pm	0.47	0.95	0.69	0.70
	Float method using coconut	1.00pm	0.45	0.99	0.82	0.75

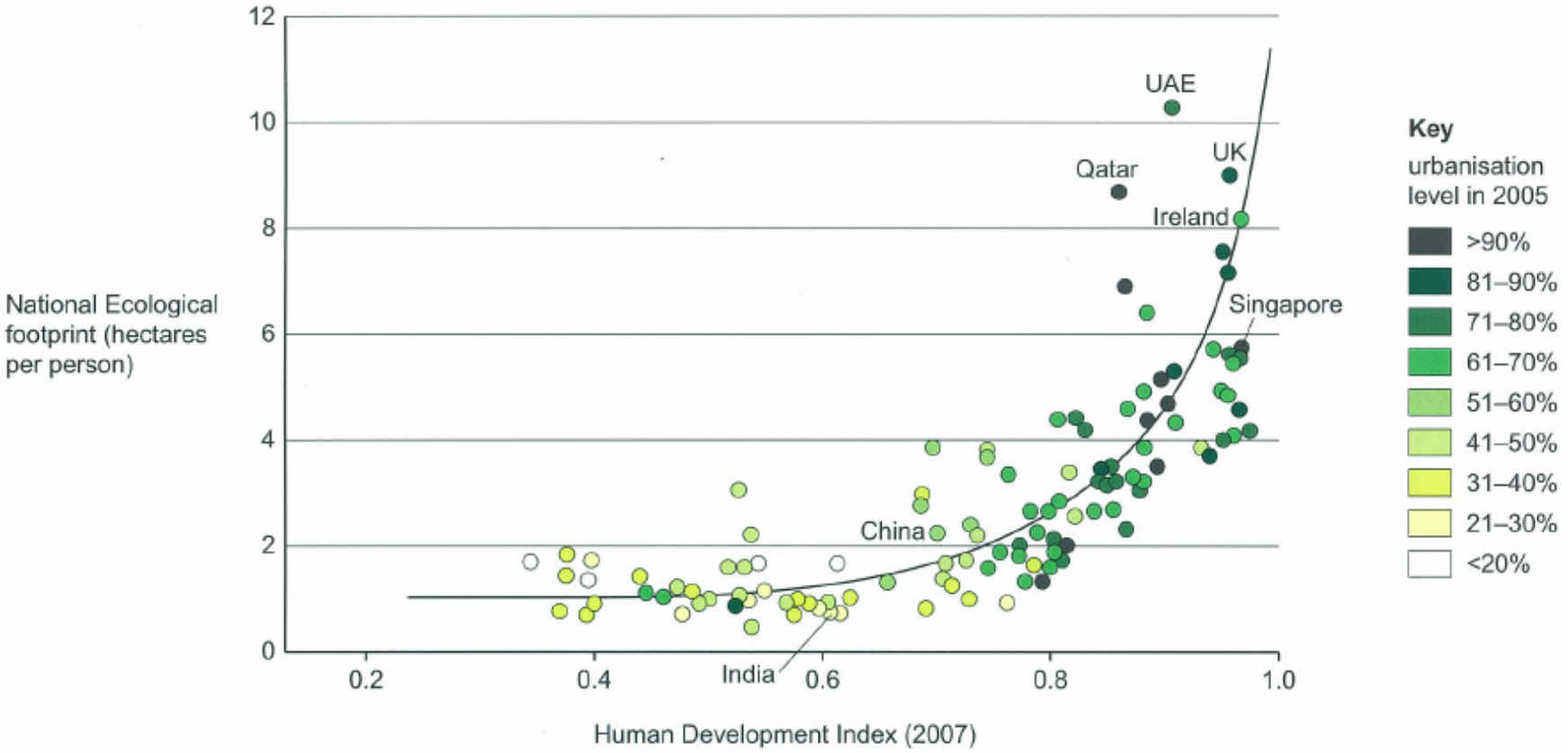
Resource 3B for Question 1

Results of cross-section measurements (width and depth) at the three sites

Distance across channel (m) from left bank		Depth of channel (m)
Site A	0.25	0.17
	0.48	0.23
	0.75	0.1
	0.93	0
Site B	0.5	0.36
	1.0	0.25
	1.5	0.07
	2.0	0
Site C	0	0
	0.9	0.32
	1.75	0.45
	2.6	0.20
	3.45	0

Resource 4 for Question 2

National Ecological footprint, Human Development Index* and urbanisation level for selected countries



**The Human Development Index is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable, and having a decent standard of living.*

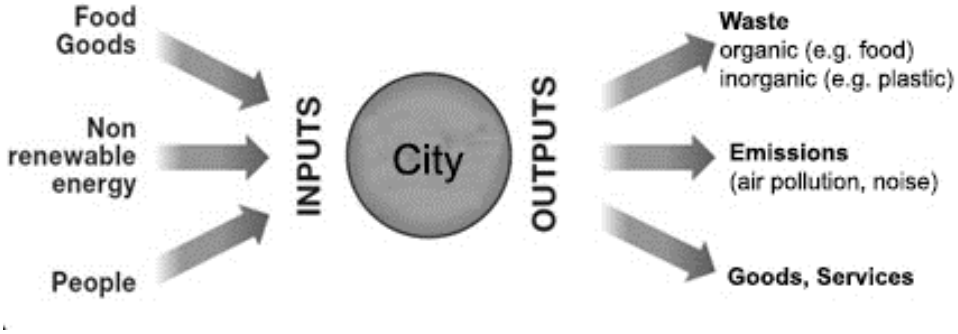
Resource 5A for Question 2

Schematic diagram of idealised urban metabolic flow[#]



Resource 5B for Question 2

Schematic diagram of typical urban metabolic flow[#]



[#]*Urban metabolic flow refers to the flows of material and energy resources and wastes in cities.*

Resource 6 for Question 2

City Development Index (CDI) for selected cities in 2010

City	Country	CDI	City housing	City infrastructure	City waste	City health	City education
Singapore	**Singapore**	94.9	91.6	99.5	100	94.8	88.6
Bangkok	**Thailand**	82.3	77.7	89.6	74.5	90.6	78.9
Hanoi	**Vietnam**	65.1	59.6	72.0	90.0	35.0	69.0
Vientiane	**Laos**	47.8	44.0	58.0	0	65.9	71.3

The UN's city development index is on a scale of 0 to 100, where 100 indicates full development.

CDI is the average of five component sub-indices of key indicators

- *City housing comprises measures of land tenure, rentals, house prices and homelessness.*
- *City infrastructure comprises access to water, sewerage, electricity and telephone.*
- *City waste comprises measures of wate water treated and formal solid waste disposal.*
- *City health comprises child mortality (under 6 years) and life expectancy.*
- *City education comprises literacy rate and a measure of school enrolment*